Johns Hopkins Engineering

Introduction to GPU Programming

Module 8 Lecture 8C NVidia GPU-Accelerated Libraries – Linear Algebra Libraries

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7. cuSPARSE
8. cuSOLVER

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8.6 cuBLAS

derived from https://developer.nvidia.com/cublas

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Intro

- The NVIDIA cuBLAS library is a fast GPU-accelerated implementation of the standard basic linear algebra subroutines (BLAS). Using cuBLAS APIs, you can speed up your applications by deploying compute-intensive operations to a single GPU or scale up and distribute work across multi-GPU configurations efficiently.
- NVBLAS is a GPU-accelerated version of BLAS that further accelerates BLAS Level-3 routines by dynamically routing BLAS calls to one or more NVIDIA GPUs as well as CPUs in the system through the cuBLAS-XT interface.
- Researchers and scientists use cuBLAS for developing gpu-accelerated algorithms in areas including high performance computing, image analysis and machine learning.

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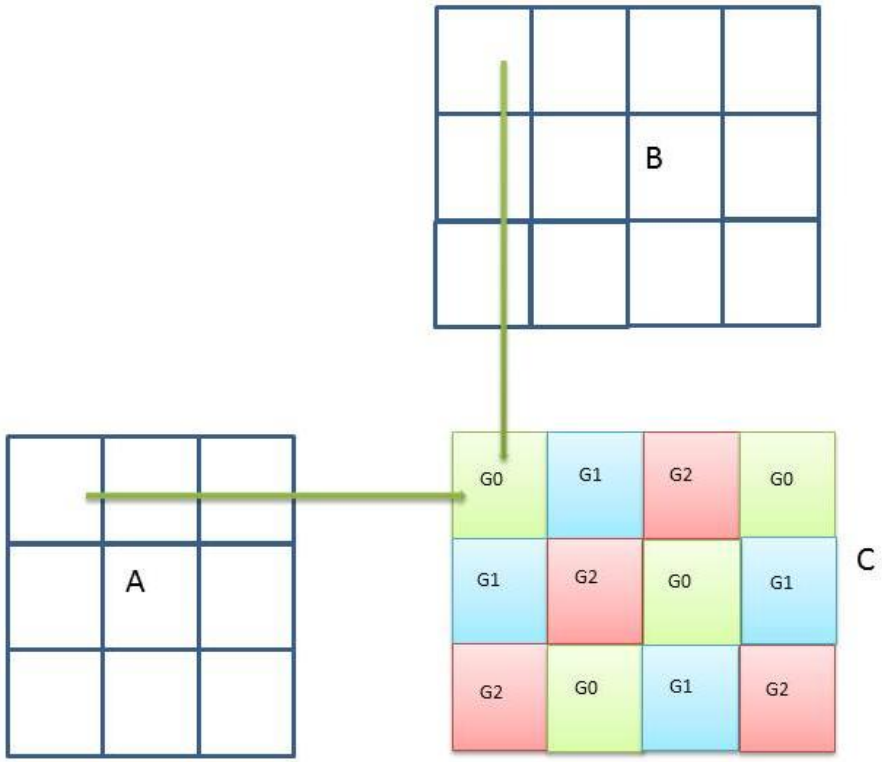
Intro



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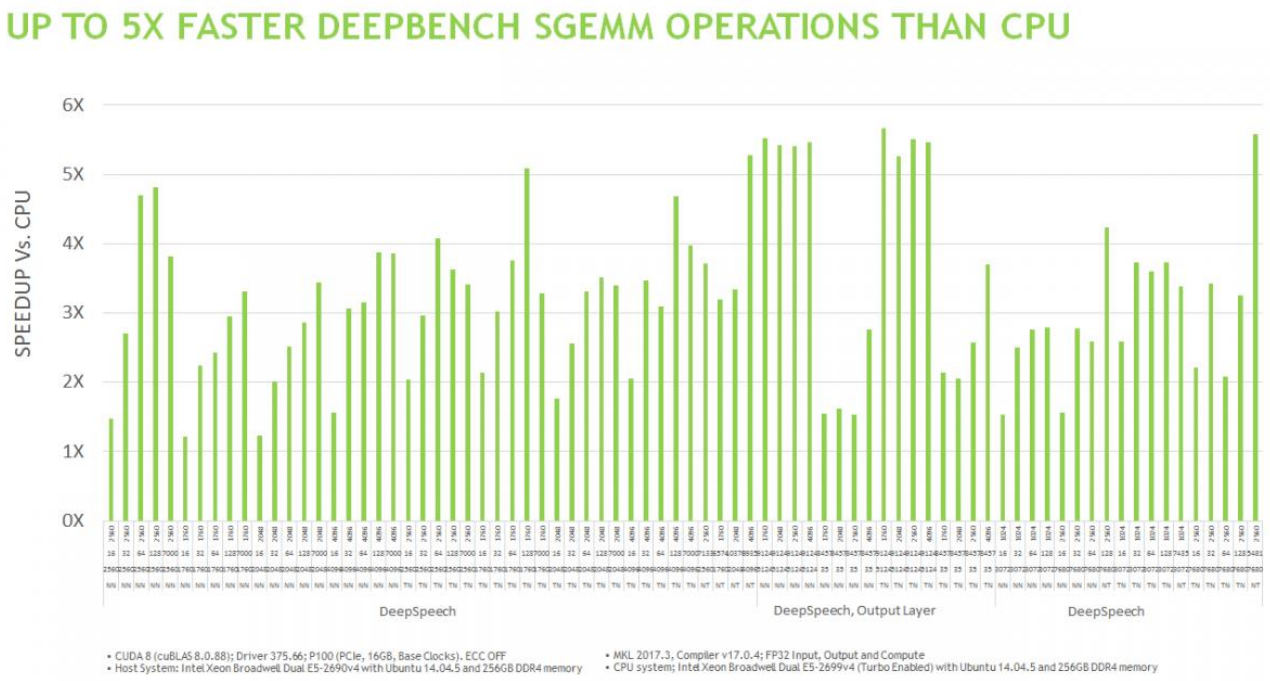
Performance



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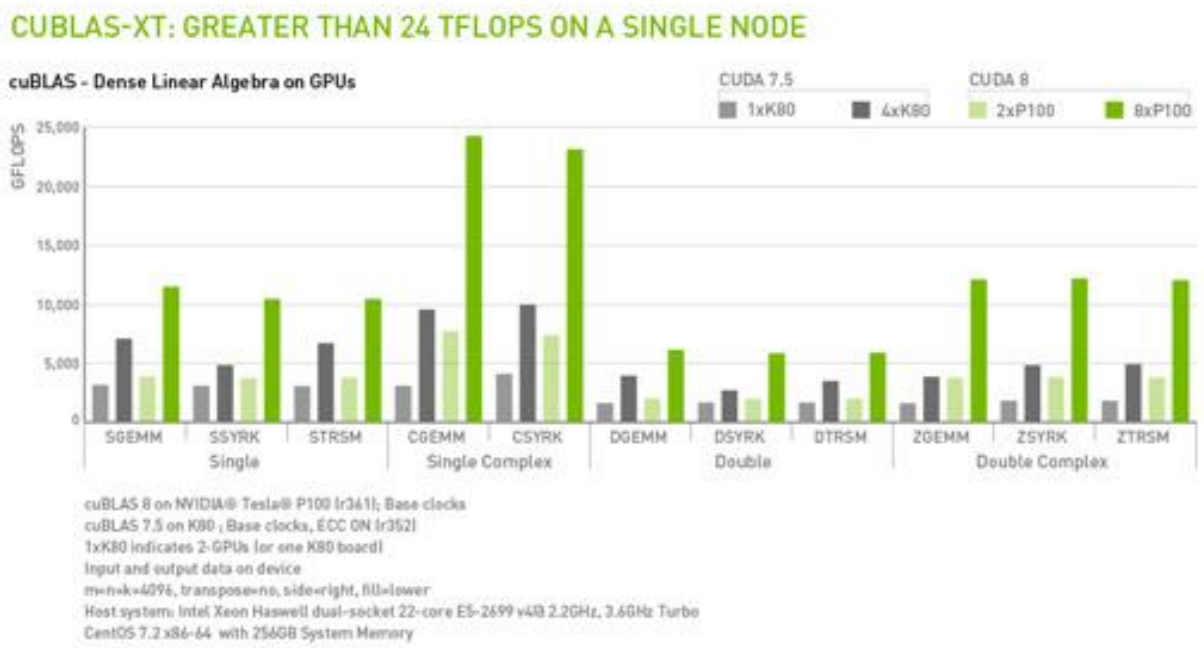
Scalability



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Key Features

- Complete support for all 152 standard BLAS routines
- Single, double, complex, and double complex data types
- Supports half-precision (FP16) and integer (INT8) matrix multiplication operations
- Support for multiple GPUs and concurrent kernels
- Supports CUDA streams for concurrent operations
- Fortran bindings
- Batch processing APIs for high performance GEMM operations, LU factorization, and matrix inverse operations
- Device API that can be called from with your own CUDA kernels
- New implementation of TRSV (Triangular solve), up to 7x faster than previous implementation

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Sample Code

```
__global__ void invokeDeviceCublasSgemm(cublasStatus_t *returnValue,
     int n, const float *d_alpha,
     const float *d_A, const float *d_B,
     const float *d_beta, float *d_C)
{
     cublasHandle_t cnpHandle;
     cublasStatus_t status = cublasCreate(&cnpHandle);
     if (status != CUBLAS_STATUS_SUCCESS) {
          *returnValue = status;
          return;
     }
      /* Perform operation using cublas */
     status = cublasSgemm(cnpHandle, CUBLAS_OP_N,
          CUBLAS_OP_N, n, n, n,d_alpha, d_A, n, d_B,
          n, d_beta, d_C, n);
     cublasDestroy(cnpHandle);
     *returnValue = status;
}
```

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8.7 cuSPARSE

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Intro

- The NVIDIA CUDA Sparse Matrix library (cuSPARSE) provides a collection of basic linear algebra subroutines used for sparse matrices that delivers up to 8x faster performance than the latest MKL. The cuSPARSE library is designed to be called from C or C++, and the latest release includes a sparse triangular solver.

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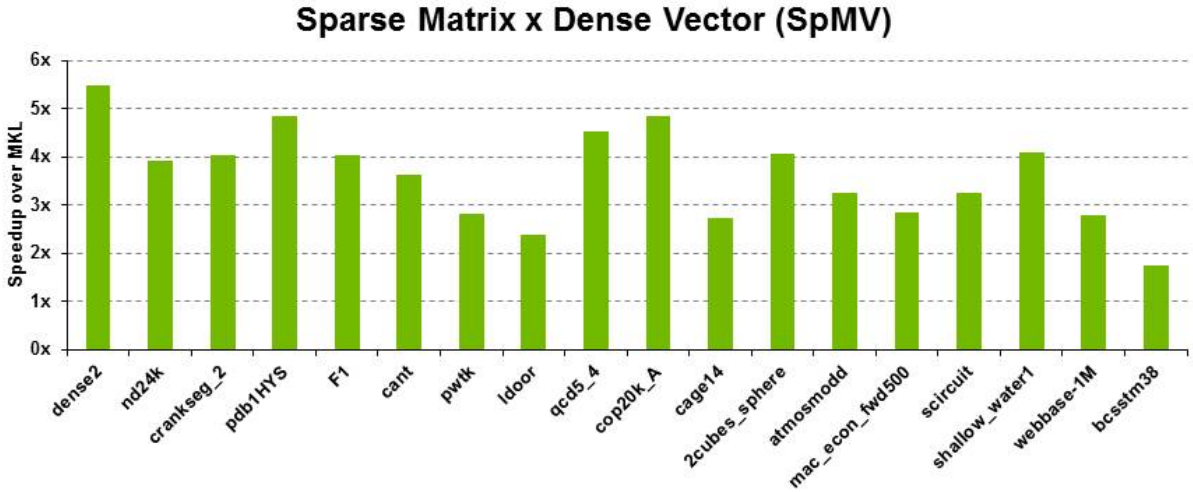
Performance

cuSPARSE: 5x Faster than MKL



- Average of s/c/d/z routines
- cuSPARSE 6.0 on K40m, ECC ON, input and output data on device
- MKL 11.0.4 on Intel IvyBridge 12-core E5-2697 v2 @ 2.70GHz
- Matrices obtained from: http://www.cise.ufl.edu/research/sparse/matrices/

Performance may vary based on OS version and motherboard configuration

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Key Features

- Supports dense, COO, CSR, CSC, ELL/HYB and Blocked CSR sparse matrix formats
- Level 1 routines for sparse vector x dense vector operations
- Level 2 routines for sparse matrix x dense vector operations
- Level 3 routines for sparse matrix x multiple dense vectors (tall matrix)
- Routines for sparse matrix by sparse matrix addition and multiplication
- Conversion routines that allow conversion between different matrix formats
- Sparse Triangular Solve
- Tri-diagonal solver
- Incomplete factorization preconditioners ilu0 and ic0

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Sample Code

```
//step 1: transform CSR to BSR with column-major order
int base, nnz;
cusparseDirection_t dirA = CUSPARSE_DIRECTION_COLUMN;
int mb = (m + blockDim-1)/blockDim; int nb = (n + blockDim-1)/blockDim;
cudaMalloc((void**)&bsrRowPtrC, sizeof(int) *(mb+1));
cusparseXcsr2bsrNnz(handle, dirA, m, n, descrA, csrRowPtrA, csrColIndA,
                                        blockDim, descrC, bsrRowPtrC);
cudaMemcpy(&nnzb, bsrRowPtrC+mb, sizeof(int), cudaMemcpyDeviceToHost);
cudaMemcpy(&base, bsrRowPtrC , sizeof(int), cudaMemcpyDeviceToHost);
nnzb -= base;
cudaMalloc((void**)&bsrColIndC, sizeof(int)*nnzb);
cudaMalloc((void**)&bsrValC, sizeof(float)*(blockDim*blockDim)*nnzb);
cusparseScsr2bsr(handle, dirA, m, n, descrA, csrValA, csrRowPtrA,
     csrColIndA, blockDim, descrC, bsrValC, bsrRowPtrC, bsrColIndC);
// step 2: allocate vector x and vector y large enough for bsrmv
cudaMalloc((void**)&x, sizeof(float)*(nb*blockDim));
cudaMalloc((void**)&y, sizeof(float)*(mb*blockDim));
cudaMemcpy(x, hx, sizeof(float)*n, cudaMemcpyHostToDevice);
cudaMemcpy(y, hy, sizeof(float)*m, cudaMemcpyHostToDevice);
// step 3: perform bsrmv
cusparseSbsrmv(handle, dirA, transA, mb, nb, alpha, descrC, bsrValC,
bsrRowPtrC, bsrColIndC, blockDim, x, beta, y);
```

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8.8 cuSOLVER

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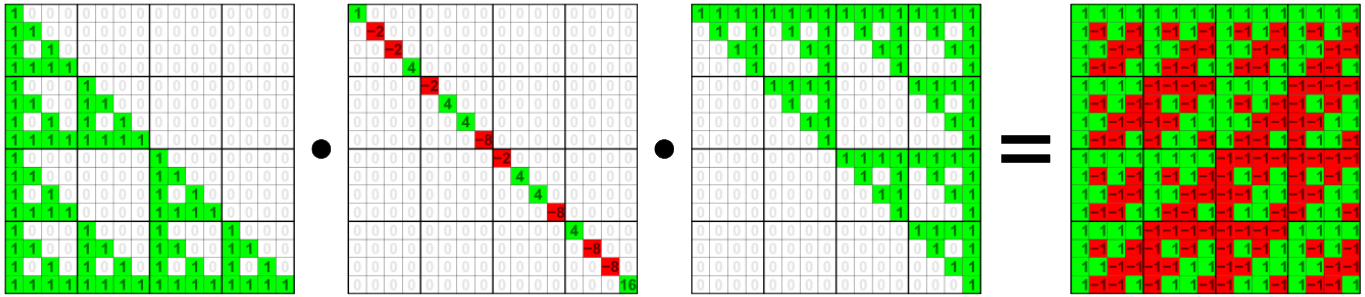
Intro

- The NVIDIA cuSOLVER library provides a collection of dense and sparse direct solvers which deliver significant acceleration for Computer Vision, CFD, Computational Chemistry, and Linear Optimization applications.



https://devblogs.nvidia.com/parallelforall/parallel-direct-solvers-with-cusolver-batched-qr/

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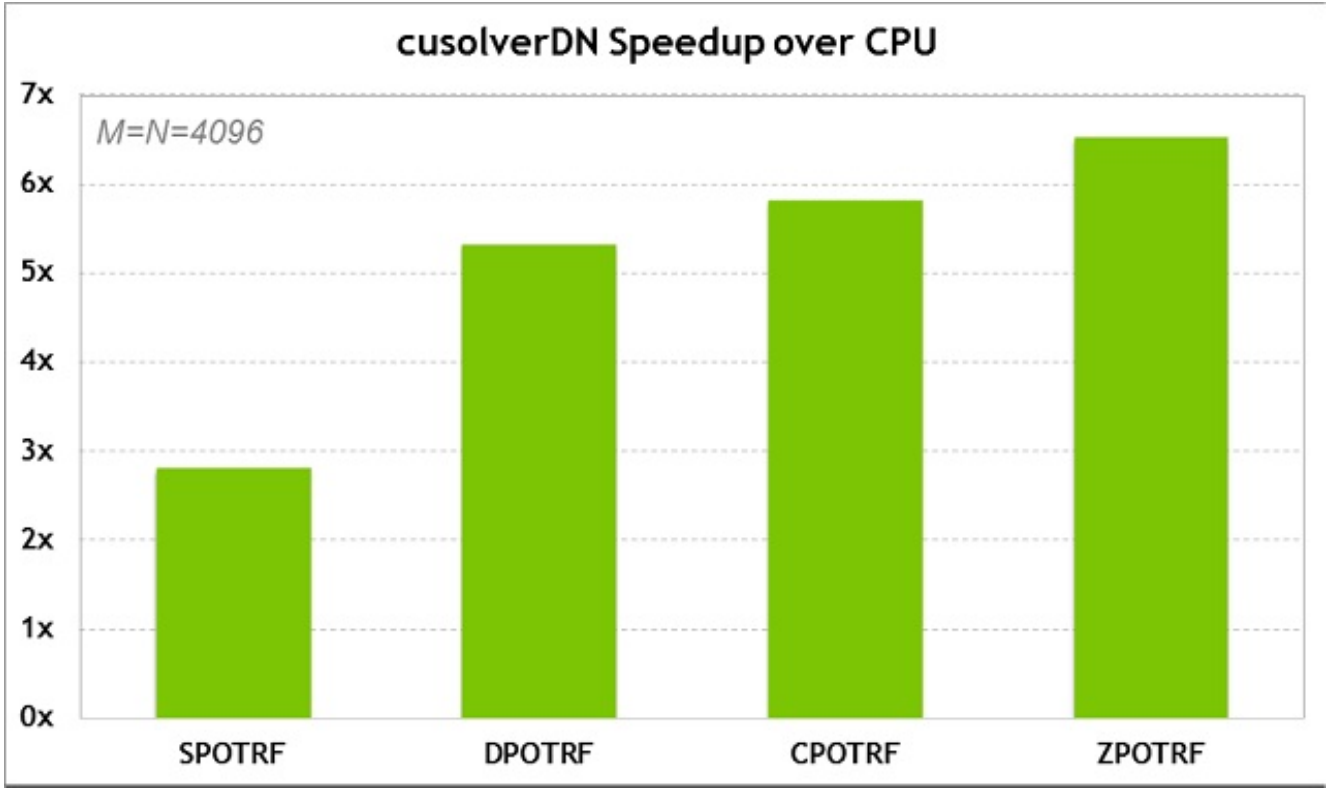

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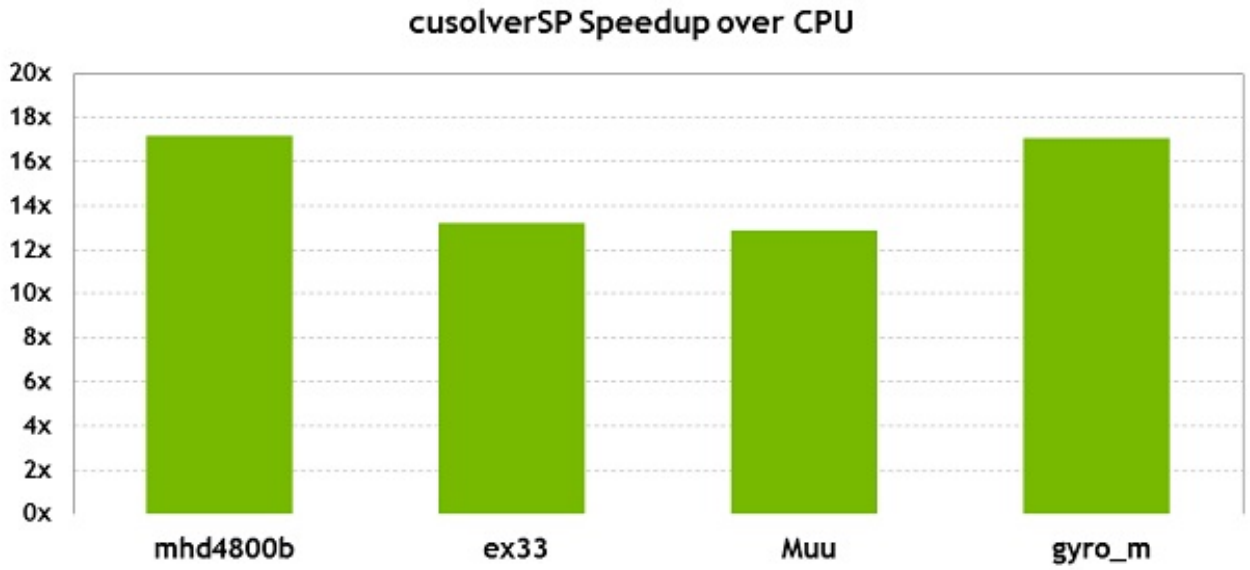

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Key Features

cusolverDN: Key LAPACK dense solvers 3-6x faster than MKL.

- Dense Cholesky, LU, SVD, QR
- Applications include: optimization, Computer Vision, CFD

cusolverSP

- Sparse direct solvers & Eigensolvers
- Applications include: Newton's method, Chemical Kinetics

cusolverRF

- Sparse refactorization solver
- Applications include: Chemistry, ODEs, Circuit simulation

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Sample Code

```
cusolver_status = cusolverSpCreate(&cusolverH);

cusparse_status = cusparseCreateMatDescr(&descrA);

cusparseSetMatType(descrA, CUSPARSE_MATRIX_TYPE_GENERAL);

cusparseSetMatIndexBase(descrA, CUSPARSE_INDEX_BASE_ONE); // base-1

cusolver_status = cusolverSpCreateCsrqrInfo(&info);

cusolver_status = cusolverSpXcsrqrAnalysisBatched( cusolverH, m, m, nnzA,
                            descrA, d_csrRowPtrA, d_csrColIndA, info);

cusolver_status = cusolverSpDcsrqrBufferInfoBatched( cusolverH, m, m,
                        nnzA, descrA, d_csrValA, d_csrRowPtrA, d_csrColIndA,
                        batchSize, info, &size_internal, &size_qr);

cusolver_status = cusolverSpDcsrqrsvBatched( cusolverH, m, m, nnzA,
                        descrA, d_csrValA, d_csrRowPtrA, d_csrColIndA, d_b,
                        d_x, batchSize, info, buffer_qr);
```